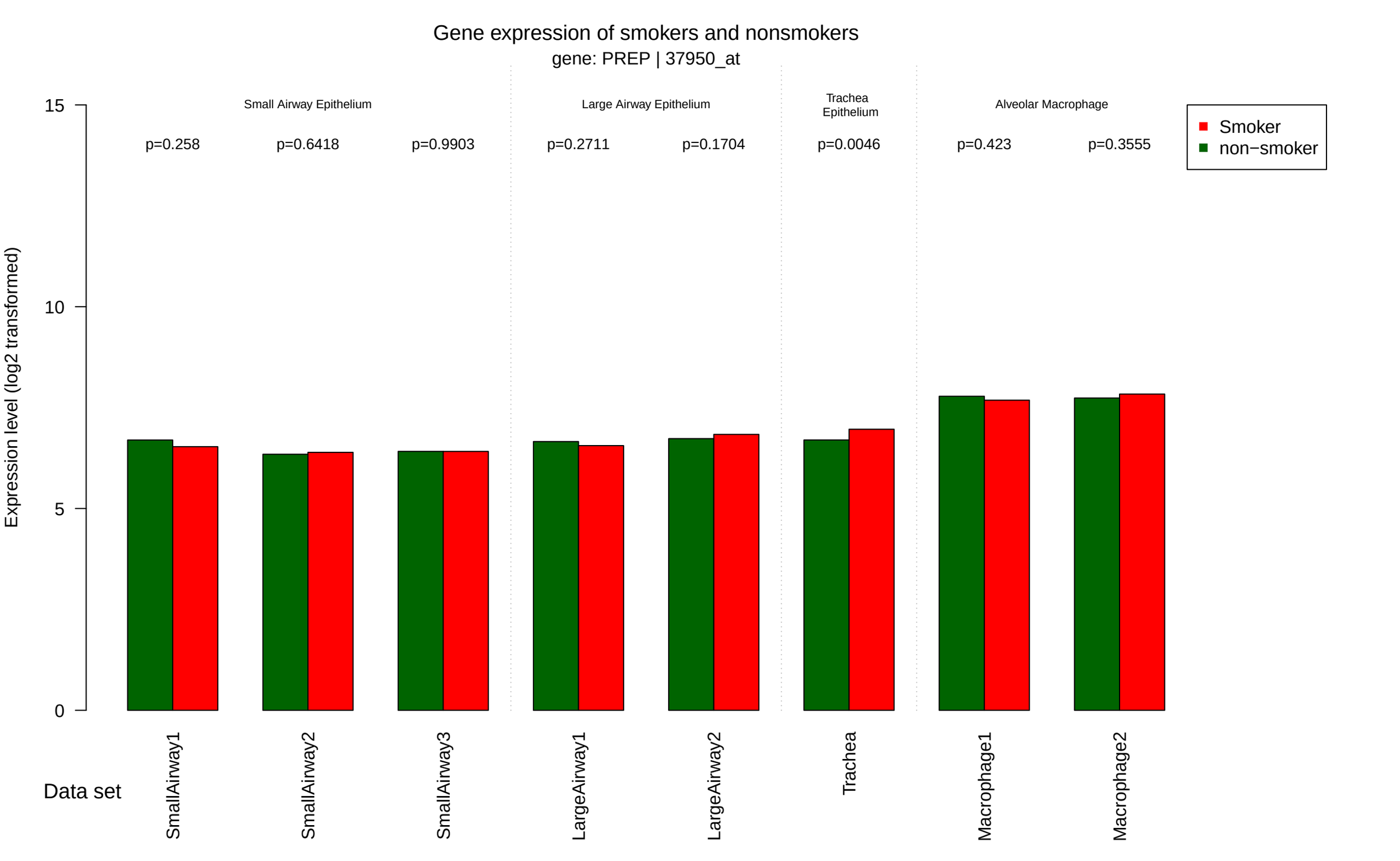
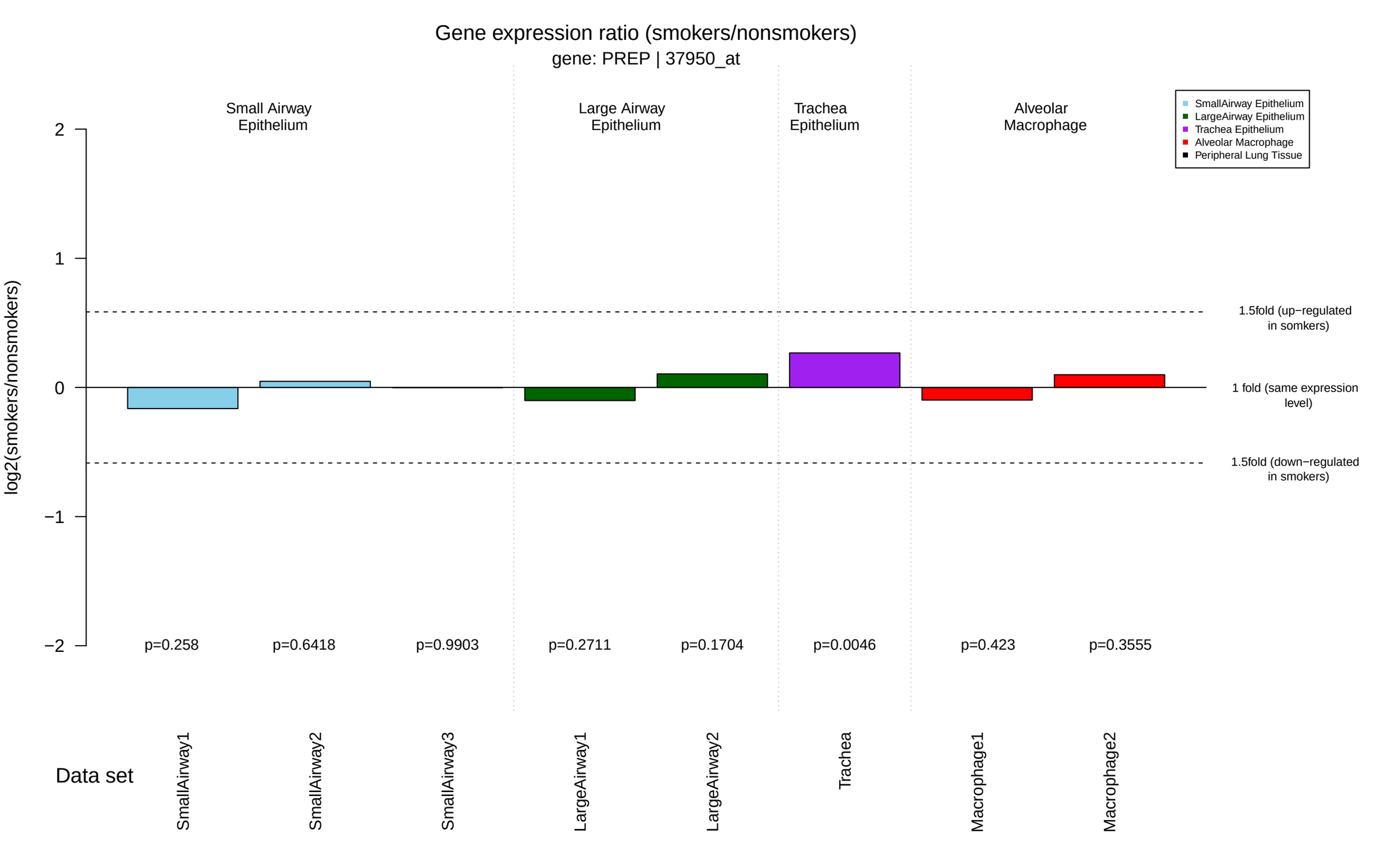
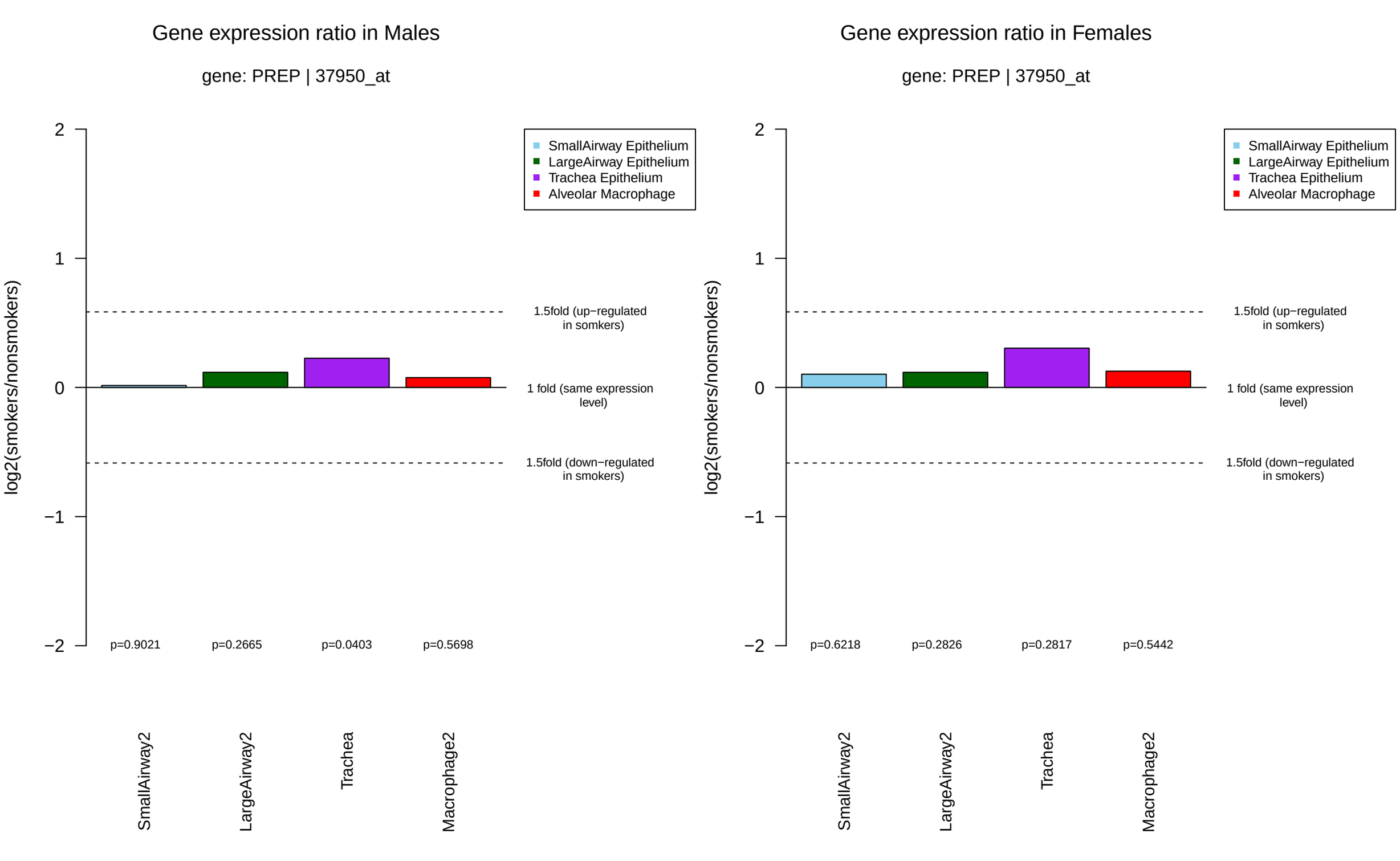
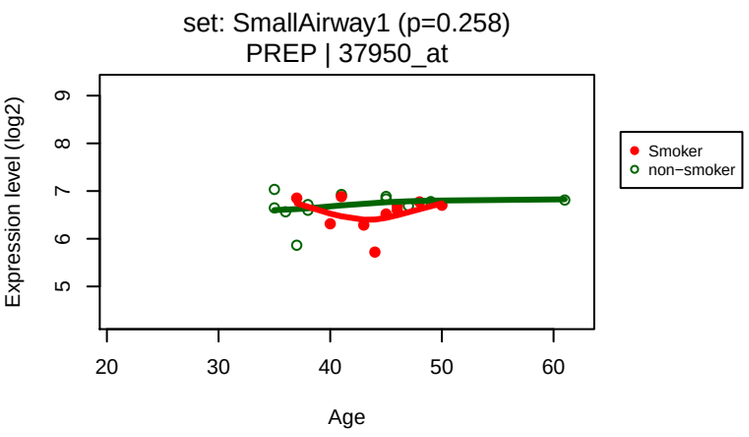
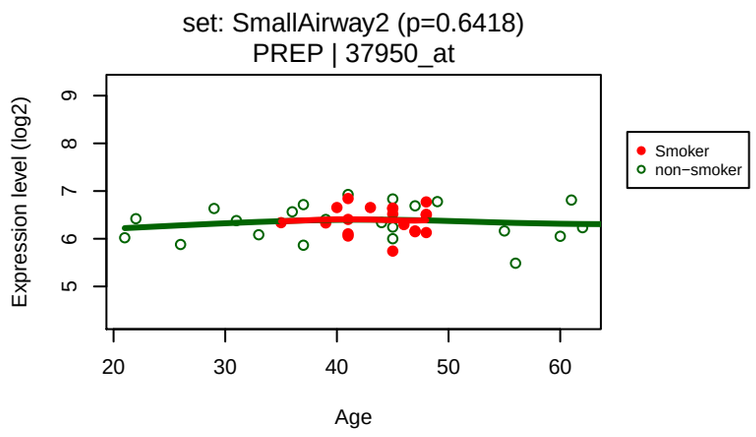
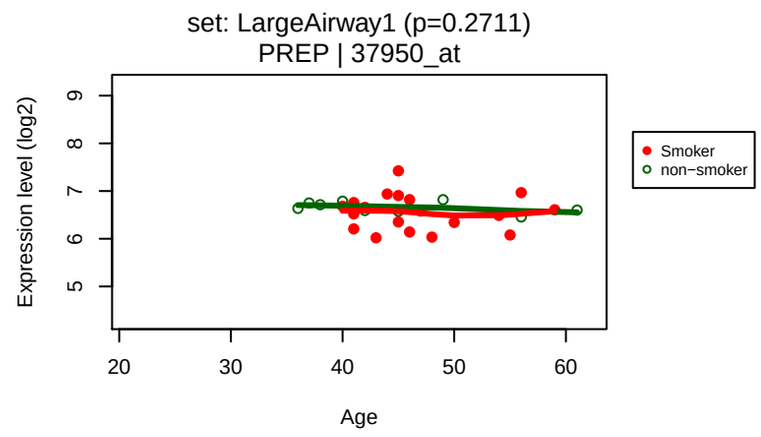

Gene expression of smokers and nonsmokers
gene: PREP | 37950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.258
p=0.6418
p=0.9903
p=0.2711
p=0.1704
p=0.0046
p=0.423
p=0.3555
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

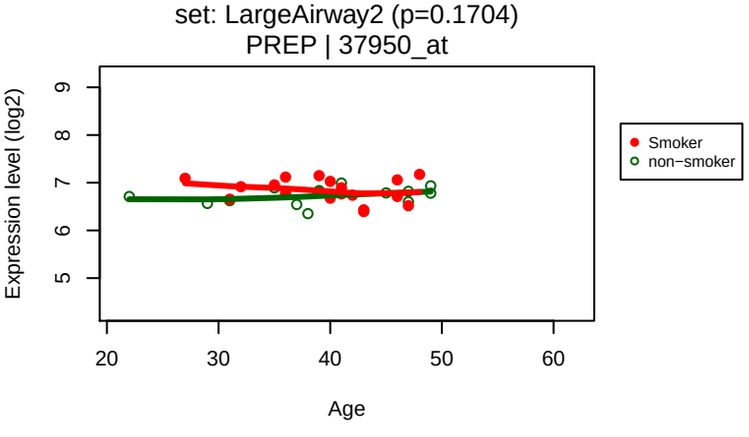

Gene expression ratio (smokers/nonsmokers)
gene: PREP | 37950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.258
p=0.6418
p=0.9903
p=0.2711
p=0.1704
p=0.0046
p=0.423
p=0.3555
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

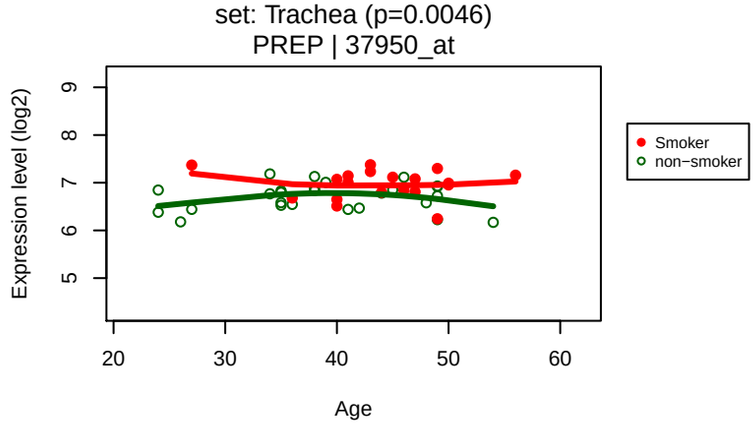

Gene expression ratio in Males
gene: PREP | 37950_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9021
p=0.2665
p=0.0403
p=0.5698
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PREP | 37950_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6218
p=0.2826
p=0.2817
p=0.5442
SmallAirway2
LargeAirway2
Trachea
Macrophage2

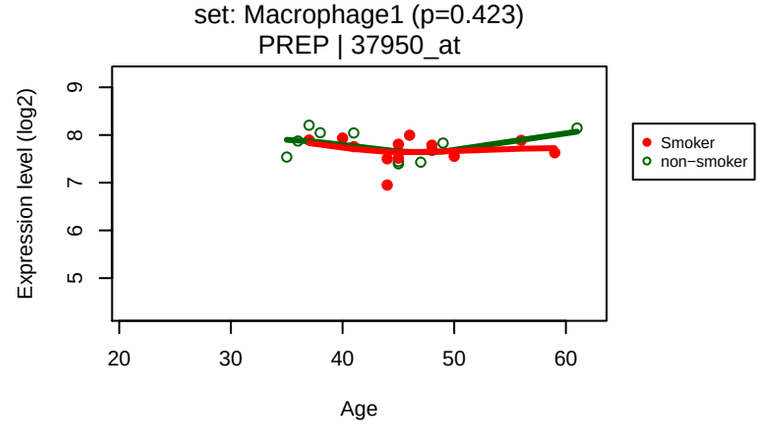

set: SmallAirway1 (p=0.258)
PREP | 37950_at
Expression level (log2)
Age
Smoker
non−smoker

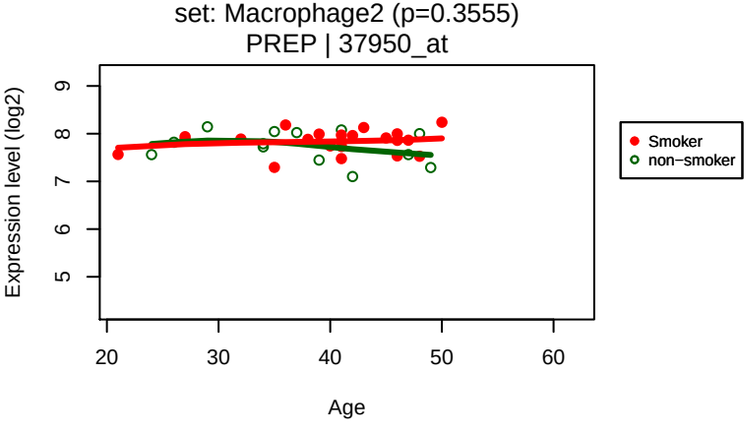

set: SmallAirway2 (p=0.6418)
PREP | 37950_at
Expression level (log2)
Age
Smoker
non−smoker

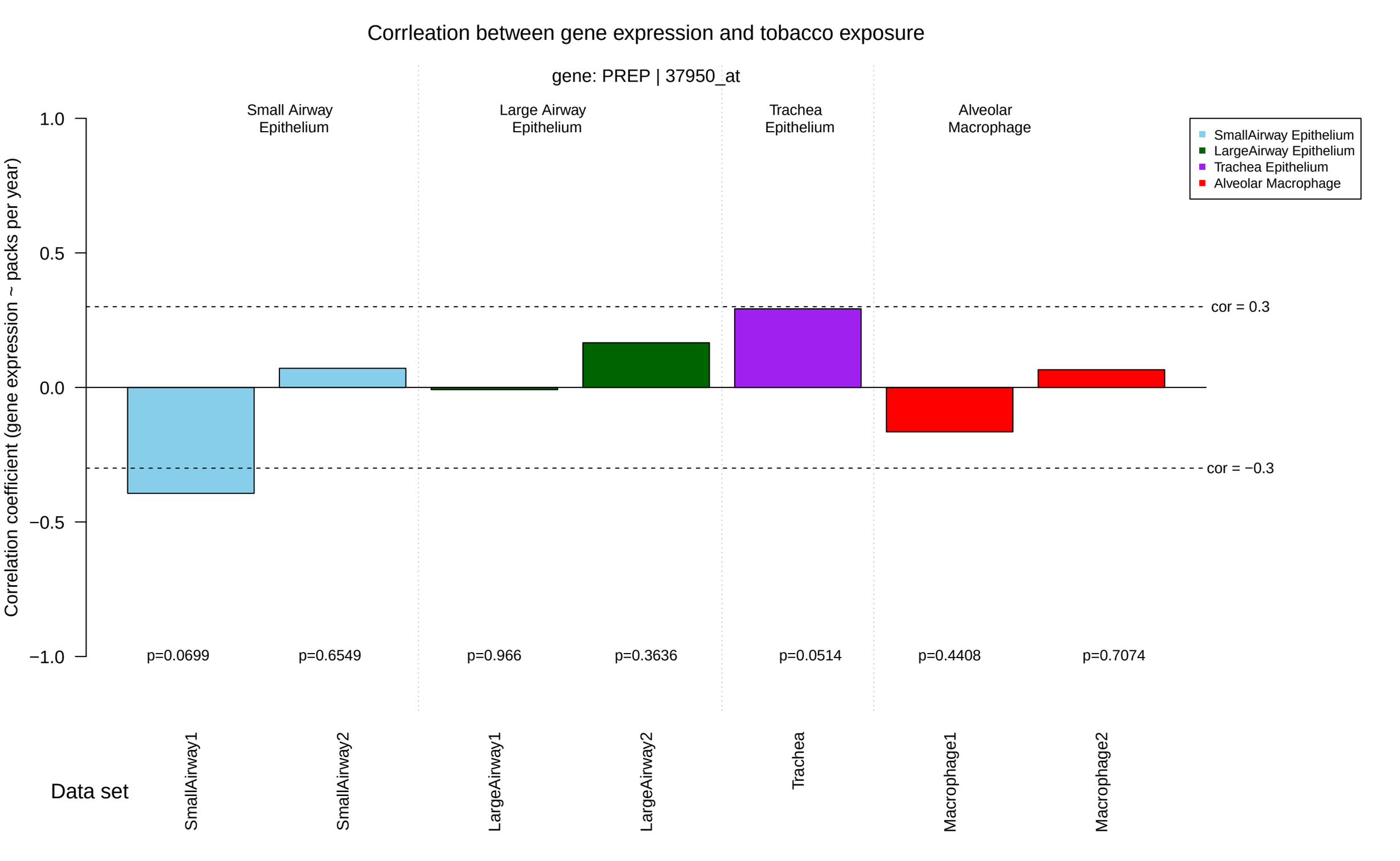

set: LargeAirway1 (p=0.2711)
PREP | 37950_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1704)
PREP | 37950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0046)
PREP | 37950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.423)
PREP | 37950_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3555)
PREP | 37950_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PREP | 37950_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0699
p=0.6549
p=0.966
p=0.3636
p=0.0514
p=0.4408
p=0.7074
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2